

Automatic Song Genre Classification in Bengali Music: A Comparative Study of Machine Learning and Deep Learning Approaches

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Abstract—Music genre categorization primarily refers to the identification of the type of music. Genre categorization is useful to create and organise our personalised playlist, including enhancing our musical experiences through music suggestions. Currently, for the purpose of categorising the genre of music, various machine learning and deep learning algorithms have been used, but it has been noticed that there is a lack of research in the classification of the Bangla music genre. Despite having some works in this field, the performance of the proposed models are not very efficient. Therefore, in this paper, we have used some leading-edge models of machine learning as well as deep learning to classify the genres of Bangla music. Six different Bangla music genres are represented in the dataset we use. Also, the dataset consists of different important features of music, such as spectral bandwidth, chroma frequency, spectral roll-off, zero crossing value, mfcc, etc. We went through intense data pre-processing, and with the assistance of a diverse range of metrics, the performance of our proposed models were assessed for multi-class classification. Moreover, it is worthy of consideration that our implemented deep neural network achieved an accuracy of about 83.65 percent.

Index Terms—Bangla Music, Genre Classification, SVM, SGD, Machine Learning, Deep Learning, MLP

I. INTRODUCTION

Bangla music comes in many different styles. It encompasses a long tradition of rich and diverse tapestry of religious and secular compositions spanning almost thousand years. It is written in Bangla language with broad spectrum of genre and music. Bangla is predicted to be spoken by 272.8 million people worldwide in 2023 [1], ranking it as the sixth most popular language in the world. Moreover, Bangla music has gained global recognition in recent years. The rise of Internet and social media have made it easier for people to discover various types of new music around the world, including Bangla music. Accordingly, it is obvious that Bangla music has been captivating a large number of audiences for centuries with its diverse range and soulful

music as well as Lyrical depth including with cultural significance.

Bangla music encompasses a diverse range of genres and subgenres, including music from Bangladesh and West Bengal, India. The number of categories and subcategories is determined by how finely we wish to classify the music. Music categories can be identified by analysing similar features and themes in songs and compositions, which allows us to classify them. In this article, we have considered some of the major genres of Bangla music. For automated music playlist generator system as well as playlist recommendation, spontaneous classification of music is needed to provide songs to listeners that correspond to their tastes and could bring them enjoyment. It is relevant to mention that, in this paper our principal objective is to categorize music genres, which will be advantageous to build an online Bangla Music platform that can suggest the audiences a wide spectrum of musical varieties in terms of their preferences.

Currently, there are numerous amounts of smart tools that helps to discover new music for other languages, but, there is a notable absence of appropriate tools for the analysis and categorization of Bangla music. In addition, diverse varieties of machine learning approaches and algorithms have been used until the present time, yet, it is necessary to train them properly to get effective result for automated music playlist generator system. Since, it is the arduous task of categorizing Bangla music, however, only a minor number of datasets have been enhanced for Bangla music genre classification music which is not well earned to fulfill the demand of providing playlist recommendation to the listeners of Bangla music. Consequently, the outcome is unsatisfactory.

We attempted to improve the performance of Music Genre Classification for Bangla Music in this paper by adopting

several machine learning and Deep Learning Methods. Our primary goal is to use simple machine learning algorithms and ensemble techniques that are computationally efficient in order to create a simpler and more robust system, as deep learning can be a resource-intensive technique. We also used dimensionality reduction to remove extraneous data from the datasets. As a result, we obtained a satisfying result that is superior to previous studies accessible and computationally cost effective.

II. RELATED WORKS

While research on music classification has gained significant traction, studies focusing specifically on Bengali music remain noticeably inadequate. Furthermore, there aren't many datasets available to properly classify Bangla music. The available works are either inefficient or computationally prohibitively expensive.

West et al., [2] how to use different portions of a song converted to audio signal for automated music classification. They emphasised the significance of segmentation-based onset identification as well as the sub-optimal nature of extracted characteristics for genre classification.

Applying k-nearest neighbor (KNN) and Support Vector Machine (SVM), Ali et al., [3] devised a machine learning-based system for classifying song genres. They employed Mel-frequency cepstral coefficients (MFCCs) extracted from audio files to capture the characteristics of each song. Despite the fact that their work is not based on Bangla music classification, their classifier performed well.

Al Mamun et al., [4] utilized deep learning libraries from tensorflow-keras and its sequential model to achieve a top accuracy of 74% in their proposed Neural Network model for Bangla music genre classification. Their deep learning model used ReLU as the activation function and the Adam optimizer. In comparison to machine learning algorithms, their model outperformed SVM (21%) and K-NN (35%) in terms of accuracy.

Ahmed et al., [5] demonstrated that their deep neural network model excels in comparison to the applied machine learning algorithms, outperforming them in terms of various performance metrics. When compared to other models, the stochastic gradient descent (SGD) exhibits the lowest accuracy. Of all the applied algorithms, random forest achieves an accuracy of 69.05%, while the SVM-RBF attains an accuracy of 68.77%. In contrast, the deep learning model demonstrates a superior accuracy of 77.68%

Sarma et al., [6] developed a dataset for Bengali music genre classification called BMGCD, comprising 2944 Bengali music clips. Introducing a Gated Recurrent Unit based deep learning model, they achieved an accuracy of 80.6% in

predicting music genres from audio signals.

Our goal will be to further investigate this dataset and devise an efficient approach based on performance metrics and computational efficiency.

III. BACKGROUND STUDY

The goal of classification is to categorise input data into predefined classes or labels. It is a type of supervised learning task. An efficient classifier's primary goals include precisely assigning the correct class labels to new, unseen instances, minimising classification errors, and generalising well to handle diverse and real-world data. An effective classifier should have high accuracy, robustness, and the ability to perform well on a variety of datasets. Furthermore, it should optimise the model for the specific needs of the application or problem at hand, balancing the trade-off between precision and recall. The algorithms studied for implementation in the project are briefly explained in the following subsections.

A. Classification using Machine Learning

SVM is one of the best machine learning classifiers available. A support vector machine is a highly effective supervised learning algorithm that is used for classification and regression tasks. It seeks the best feasible line or decision boundaries between data points from various classes, with the goal of maximising the margin between the hyperplane and the closest data points of each class. This facilitates the distinction of data classes and the attainment of accurate classification and regression results [7]. SVM is an excellent classification algorithm. It is capable of handling high-dimensional data. It also works well with small datasets. Furthermore, kernels and hyperplanes play important roles in data point classification. A kernel function is a method for transforming input data into the format required for processing. It enables SVMs to operate in a higher-dimensional space, allowing the separation of non-linearly separable data points by transforming them into a higher-dimensional linear equation. SVMs employ a variety of kernel functions, including linear, nonlinear, polynomial, radial basis function (RBF), and sigmoid kernels. Each type of kernel function has unique properties and applications [8]. A hyperplane is a decision boundary used in SVM to separate data points of different classes. SVM seeks a hyperplane that maximises the margin, or the maximum distance between data points of different classes [9]. Because of their efficiency and dependability, the linear kernel and rbf kernel seemed most appropriate for our article.

The k-nearest neighbours (KNN) algorithm is a supervised learning classifier that is non-parametric in nature. It predicts the classification of a new sample point based on the similarity measure of previously stored data points using data from multiple classes. KNN classifies new data points based on their neighbours' similarity, and k represents the number of nearest neighbours used to classify new data points [10] [11].

Stochastic Gradient Descent (SGD) is an optimisation approach used in machine learning to train models efficiently. It is a gradient descent variation that processes training data in small batches or individual data points rather than the full dataset at once, making it computationally efficient and ideal for large-scale datasets [12].

Both the decision tree and the random forest algorithms are used to solve classification and regression problems. Based on regression and classification types, the decision tree generates a series of decisions that are used to implicate specific results. Random forest, on the other hand, combines several decision trees to make a prediction. The primary distinction between the two is that decision trees are simple and easy to understand, whereas random forests are more complex and difficult to deconstruct. They do, however, necessitate extensive training and can be slow, particularly for real-time predictions. Decision trees, on the other hand, are quick and easy to use on large datasets, particularly linear ones. [13]

B. Classification using Neural Networks

Deep Neural Networks (DNNs) are a type of machine learning technique that is comparable to artificial neural networks. It seeks to emulate human brain information processing and is especially useful for dealing with unlabeled and unstructured data. DNNs are made up of multiple layers, typically two or more, with input, output, and at least one hidden layer in between. They have become the industry standard for addressing numerous computer vision tasks and are frequently used to handle difficult problems that simpler models cannot answer. DNNs, like shallow artificial neural networks, can simulate complicated non-linear interactions and are commonly employed in supervised and reinforcement learning issues. They are especially useful for picture classification, time series analysis, and text analysis [14]. On the other hand, A Multilayer Perceptron (MLP) is a form of artificial neural network that consists of fully linked neurons with a nonlinear activation function. It consists of three or more layers: an input layer, an output layer, and one or more hidden layers. The MLP is trained via backpropagation to learn the relationship between linear and nonlinear data. It is a standard feedforward artificial neural network example [15].

IV. DATASET

This project relied on the use of the "Bangla Music Dataset" dataset [16]. There are roughly 1742 songs total, divided into six categories. The groups consist of Adunik Music, Hip-Hop Music, Band Music, Nazrulgeeti (Songs written and composed by Kazi Nazrul Islam), Folk song which is known as 'Palligeeti', Rabindra Sangeet(Songs written and composed by Rabindranath Tagore). Every song has thirty features altogether. RMS, zero crossing, spectral centroid, spectral

rolloff, spectral bandwidth, chroma frequency, melspectrogram, mfcc, etc. are typically a song's primary characteristics. These characteristics are specific to each song, but they read similarly across every category. The amount of songs in each category is displayed in a Bar graph in Figure 1. Additionally, Figure 2 shows a pie chart of the data dissemination.

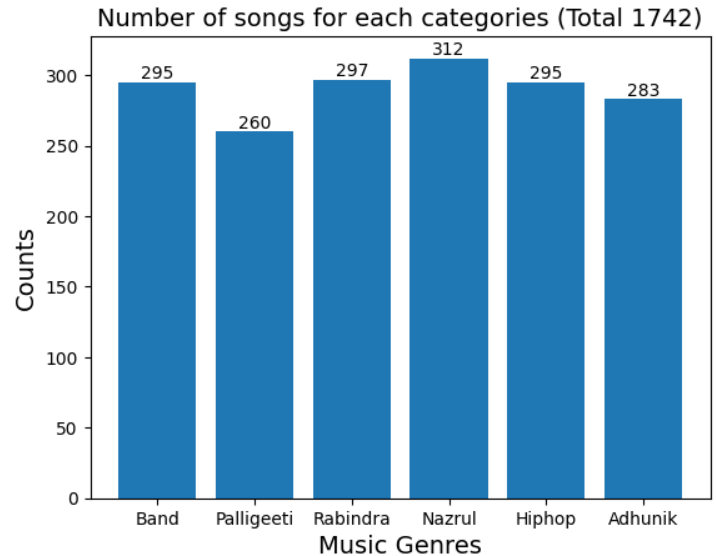


Fig. 1. Number of songs in each Genres.

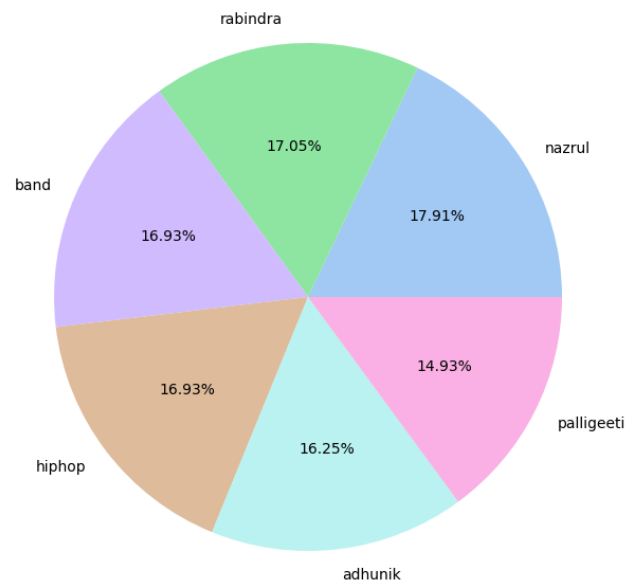


Fig. 2. Data Distribution of Each Categories.

V. PROPOSED METHODOLOGY

Our proposed methodology is divided into two parts as shown in Figure 3: data pre-processing and implementation

of algorithms. The results are then analysed to evaluate the performance of various classifiers.

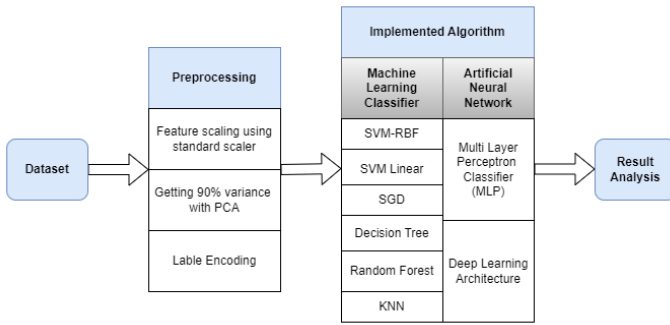


Fig. 3. Flowchart of Proposed Methodology.

A. Data Preprocessing

1) *Dimentionality Reduction*: Our dataset contains a total of twenty-nine features for each song, which is a lot of data to work with. Furthermore, some features have a greater impact than others in obtaining the target classes. That is why Dimentionality Reduction is used to remove features that have little impact on the calculation, making it easier to compute. This also makes classifiers more efficient because it eliminates the need for some unnecessary computations. Principal Component Analysis is a popular technique for reducing dimension. It is employed in order to reduce the number of features in a dataset while retaining as much information as possible. In our case, we used PCA to calculate the effects of various features. Following that, we discovered that we only needed 16 features to retain 90% of the variance data. As a result, only the most important 16 features were considered, and the rest were eliminated. As a result, the number of computations decreased, and classifiers improved in calculation and performance.

2) *Feature Scaling*: Feature Scaling is a critical step in machine learning that involves normalising a dataset's range of features. Real-world datasets frequently contain features with varying magnitude, range, and units, which can lead to inaccurate predictions by models. Scaling the features to the same scale allows the model to give equal weight to all features and make accurate predictions. Scaling can also help the optimisation process by smoothing the gradient descent flow and allowing algorithms to converge faster. Normalisation and standardisation are two scaling techniques that are used to rescale each feature to have a standard deviation of 1 and a mean of 0 [17].

3) *Label Encoding*: Label encoding is a vital preprocessing step in machine learning that converts categorical columns into numerical ones, enabling the use of algorithms that only process numerical data. By assigning unique numerical labels to distinct categories, label encoding helps machine

learning models to interpret and process categorical data more effectively [18].

B. Implementation of algorithms

There are numerous classifiers available today to achieve efficient results in classification problems. To achieve our desired classification results, we used six machine learning classifiers and two neural network-based approaches. Also, the hyperparameters were tuned accordingly.

1) *Machine Learning Algorithms*: We tried both the linear kernel and the radial basis function kernel for SVM. C, which denotes the regularisation parameter, was set to 0.025 for the linear kernel. The hyperparameters max depth=10, random state=42, max features=None, and min samples leaf=15 were set for the decision tree. The loss was set to 'modified huber' for SGD. Random forest and K-nearest neighbours were also used.

2) *Artificial Neural Networks*: In the case of artificial neural network approaches, we started with a Multi Layer Perceptron classifier. The activation function for MLP was 'relu,' the solver or optimizer was 'adam,' the initial learning rate was 0.001, and the maximum iteration was 1000 with early stopping enabled. A sequential model with six dense layers was used for our custom neural network architecture. The number of neurons was 256, 128, 64, 32, 16, and 6 in that order. The activation function was 'relu' for all layers except the last, which used the 'sigmoid' function. Also, the dropout values for the layers were 0.6, 0.4, 0.3, and 0.2, respectively. The loss was set to sparse categorical crossentropy during compilation, and the optimizer was rmsprop. In addition, 150 epochs were used with an 8-batch size and early stopping enabled.

VI. EXPERIMENTAL RESULT ANALYSIS

A. Experimental Tool

We used Anaconda distribution and Python 3.10 as the programming language for our project. Python library functions were used for programming, and sci-kit learn was used to implement machine learning algorithms. Keras library was used for implementing deep neural networks. For our research, we used a computer with 8 gigabytes of RAM, a core i5-1235U CPU, and an RTX 3050 GPU. The GPU was used to train the deep neural network model.

B. Results

The evaluation criteria of recall, precision, f1-score, and accuracy were determined for each model under consideration. These metrics are standard tools for performance evaluation. We implemented various machine learning classifiers as well as artificial neural network. With regard to machine learning classifiers, we have used Random forest, SVM-RBF, SVM

Linear, SGD, Decision tree and KNN. Additionally, in the context of artificial neural network, we have implemented Multi Layer Perceptron classifier and our custom Deep Learning architecture. The accuracy, precision, recall, f1-score across all our models in our analysis has been illustrated in Table I.

TABLE I
PERFORMANCE COMPARISON OF IMPLEMENTED MODELS

Method	Accuracy	Precision	Recall	F1 Score
Random Forest	71.22	71.66	71.66	71.07
SVM-RBF	74.35	74.63	74.37	74.24
SVM-Linear	71.52	71.10	71.47	71.29
SGD	64.78	65.00	64.81	64.18
Decision Tree	61.26	61.59	61.52	61.03
KNN	65.57	70.10	65.41	65.32
MLP	75.61	75.65	75.85	75.48
Deep Neural Network	83.65	83.92	83.27	83.11

When compared with machine learning classifiers and our deep learning architecture, the deep learning architecture has achieved the highest performance which is 83.65%. Among all the implemented machine learning classifiers, it is evident that SVM-RBF excelled with an accuracy of 74.35%. Moreover, SVM-Linear and Random Forest exhibited almost similar levels of accuracy, both recording 71.52% and 71.22%, respectively, closely approaching the accuracy of SVM-RBF. Furthermore, SGD, Decision tree, KNN has achieved the accuracy of 64.78%, 61.26%, 65.57% respectively. The results show significant improvement from other related works [6] [5] both in performance of machine learning classifiers and neural network classifiers. Our revised neural network architecture outperforms all classifiers developed for this project as well as networks developed by other researchers [5] [6]. The accuracy of the neural network architecture proposed by Ahmed et al., [5] was about 77.68% and the Gated Recurrent Unit based model proposed by Sarma et al., [6] was about 80.6% whereas our neural network architecture was able to achieve a outstanding accuracy of 83.65% while also being computationally cost effective. The dimensionality reduction is one of the primary reasons why our proposed method works relatively better. Even though the dataset is not very large, it has an extensive amount of dimensions, making it difficult for classifiers to find enough characteristics for each class. Our dimensionality reduction technique reduced the dimensions, leaving only the most important features. This allows the classifiers to work more efficiently and provide better performance. In addition, our neural network is one layer larger than the model proposed by Ahmed et al., [5], and as our dimension was reduced, it also became more computationally more efficient. It could also handle the data with great effectiveness.

VII. CONCLUSION

The classification of Bangla music genres is useful for developing any dedicated online-based Bangla music platform. Furthermore, it is beneficial to organise all music in terms of category to assist any individual in finding preferable music in any specific online Bangla Music platform. Despite

achieving excellent results, all of our models can be further trained and made more efficient. One approach is to gradually tune the hyperparameters more. In addition, obtaining a larger dataset can also be a breakthrough. We intend to improve this dataset by adding more songs with correct features and using it in future research to improve overall performance and make bengali music genre classification more efficient.

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